

How to tell a **real** finding from a good story

Everything in this course is one question asked over and over:

somebody claims X causes Y — should you believe them? The

readings are not really about Kansas, welfare, or crime. They are

fourteen worked examples of that single question. Learn the

machinery below and every reading becomes predictable.

Also available as audio — a 44-minute condensed edition and an

85-minute full one, with a spoken self-test. Timestamps.

WHAT THE GRADE IS MADE OF

Final — cumulative, Dec 17	35%
Midterm — in class, Oct 15	30%
Six quizzes — 5% each	30%
Attendance — Top Hat, code 782449	5%

Read every assignment twice: once for the argument, once for the move.

Trounstone pairs a popular claim with a scholar who tests it. Frank says Kansas votes against its interests; Bartels checks the survey data. Bloomberg says stop-and-frisk isn't profiling; Bertrand & Mullainathan randomize names on résumés. Gladwell says crime is an epidemic; Levitt does the arithmetic. **The pairing is the lesson.** When you read the journalism, ask what evidence would settle it. When you read the scholar, name the technique.

Fourteen concepts · seven debates · one recurring exam question

Printable edition. Wide margins are left on purpose — mark it up. The interactive version with hover values on every chart is in your Claude artifacts, and the two audio editions were delivered in the conversation.

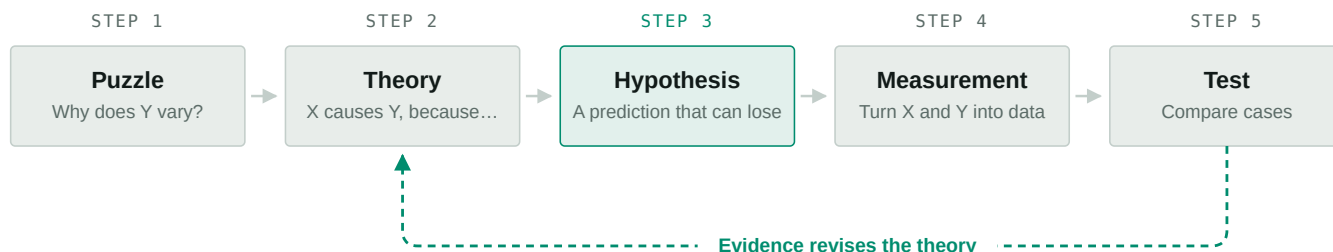
UNIT 1

Theory-driven research

Lectures 1–2 · Aug 27 & Sep 1 · Pollock Intro & Ch. 1 · Bueno de Mesquita & Fowler Ch. 3 · Fisman, Gelman & Stephenson

Science isn't a pile of facts. It's a **loop**: you have a hunch about why something happens, you turn the hunch into a prediction that could turn out wrong, you collect evidence, and you find out. The part that makes it science is that **the prediction can lose**.

FIGURE 1 · THE RESEARCH LOOP



The loop is the whole course. Notice where the pressure sits: Step 3 (can this prediction lose?) and Step 4 (did you measure the thing you claim to measure?). Almost every criticism in every assigned reading is aimed at one of those two boxes.

Theory	A general explanation of <i>why</i> X causes Y. It has to apply beyond the case in front of you. "Voting is partly social – people vote to avoid looking like a shirker."
Hypothesis	One specific, testable prediction that follows from the theory. Must name two variables and a direction. "Households told their neighbors will see their turnout record will vote at higher rates than households that aren't told."
Falsifiability	If no possible evidence could prove you wrong, you are not doing science. "Politics is complicated" is unfalsifiable.
Empirical vs. normative	Empirical = what <i>is</i> , settled by evidence. Normative = what <i>ought</i> to be, settled by values. "Turnout is 60%" vs. "turnout should be higher."

Where numbers come from — the first reading of the semester

Fisman, Gelman & Stephenson trace famous statistics back to their sources and find most of them have none. Their examples are quiz-friendly:

- "You lose 80% of body heat through your head" — from a 1950s military study where subjects wore Arctic suits covering everything *but* the head. The real figure is roughly **10%**.
- "Bribery costs \$1 trillion a year" — the midpoint of a **\$600 billion–\$1.7 trillion** range from a handful of old surveys. Taking the midpoint deletes the uncertainty, which was the most honest part.
- "Corruption costs \$2.6 trillion" — traces to a one-sentence bullet in an advocacy brief that **cited no source**, then got repeated by the heads of the UN and OECD.
- Dan Quayle's "70% of the world's lawyers are American" — the real share is about **25–35%**, in line with the U.S. share of world GDP.

THEIR FOUR RULES – MEMORIZE THESE

1. Trace the number to the original source, not to whoever cited it. "Studies have shown" and "it has been estimated" are red flags. **2.** Watch for **statistics laundering**: a shaky figure gets repeated by a respectable body, then everyone cites the respectable body. **3.** A Harvard professor's guess is not a "Harvard estimate." **4.** Beware the broken-telephone effect — ranges collapse to points, caveats fall off, narrow findings get generalized.

UNIT 2 Concepts and measurement

Lecture 4 · Sep 8 · Pollock Ch. 2–3 · plus the whole Kansas debate, Lecture 5

You cannot test a claim about "the working class" until you say exactly who counts. Measurement is where most political arguments actually go wrong — and where this course spends the most time.

Concept	The abstract idea you care about. democracy · ideology · working class · discrimination
Conceptual definition	What you mean by the concept, in words. Narrows the idea. "Working class = people whose income depends on wages rather than assets or credentials."
Operational definition	The exact recipe for measuring it — specific enough that a stranger would produce the same numbers. "Working class = white respondents with no four-year college degree."
Variable / value	The measured thing that varies across cases; a value is what one case scores on it.
Unit of analysis	What each row of your data is: a person, a county, a country, a bill, a résumé.

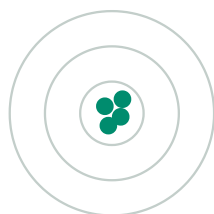
Validity

Are you measuring the thing you claim to measure? A *bias* problem.

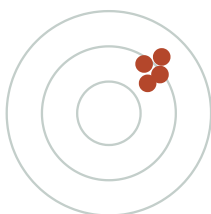
Reliability

Does the same procedure give the same answer when repeated? A *consistency* problem.

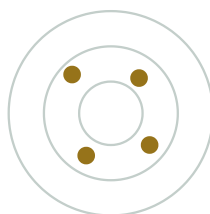
FIGURE 2 · VALIDITY VS. RELIABILITY



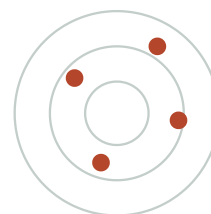
Valid + reliable
On target, every time



Reliable, not valid
Consistently wrong —
a biased measure



Valid, not reliable
Right on average,
but noisy



Neither
Wrong and noisy

The dangerous box is the second one. A measure can be perfectly consistent and perfectly wrong — self-reported turnout is reliable (people answer the same way every time) but invalid (they overreport). More data never fixes a validity problem.

Levels of measurement

LEVEL	WHAT IT DOES	EXAMPLE	WHAT YOU MAY COMPUTE
Nominal	Names categories. No order.	Party ID, religion, state	Mode, percentages
Ordinal	Ranks categories, but the gaps aren't equal.	Strongly agree → strongly disagree; education level	Mode, median
Interval / ratio	Real numbers with equal gaps.	Turnout %, income, vote share, age	Mode, median, mean, standard deviation

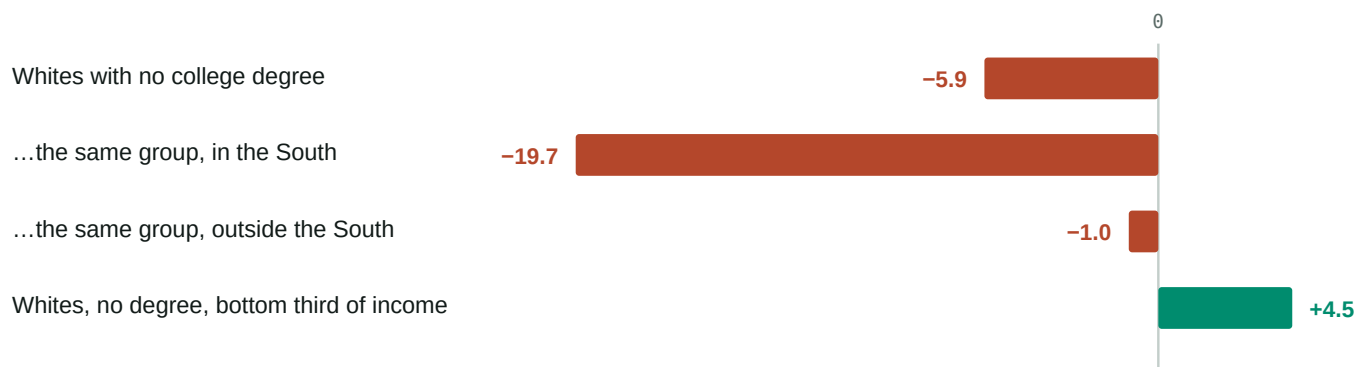
CLASSIC QUIZ TRAP

You cannot take a mean of a nominal variable. "The average religion" is nonsense. Ordinal variables get a median, not a mean — the distance from "agree" to "strongly agree" isn't a known quantity.

Why this unit decides the Kansas fight

Thomas Frank argues that white working-class voters have abandoned the Democrats because culture outweighs economics — his headline evidence is that the poorest county in America went for Bush by over 75%. Larry Bartels asks the measurement question instead: *who is the working class?* Change the operational definition and the answer flips.

FIGURE 3 · ONE CONCEPT, THREE OPERATIONAL DEFINITIONS, THREE ANSWERS



Cumulative change in the Democratic share of the two-party presidential vote, 1952–2004. Negative = moved Republican.

Bartels (2006), from 14 elections of American National Election Studies data. The national drop is real but small, and it is almost entirely Southern — outside the South, fourteen elections and fifty-two years produced a one-point shift. Define "working class" by *income* instead of education and the poorest white voters moved *toward* the Democrats. Frank never fixed his definition, so his thesis was never testable.

THE THREE NUMBERS TO REMEMBER HERE

Among white non-college voters in 2004, **40% had family incomes over \$60,000** and more than half called themselves middle class — so "no college degree" is a poor operationalization of "working class." Bush's 2004 margin among the **bottom third by income** of that group was under **two points**. And by 2020 (Ruffini) the income divide had nearly vanished: Clinton won the lowest earners by **31 points** in 1996 while Dole won the highest by 16 — a 47-point spread — versus roughly **8 points** in 2020.

UNIT 3

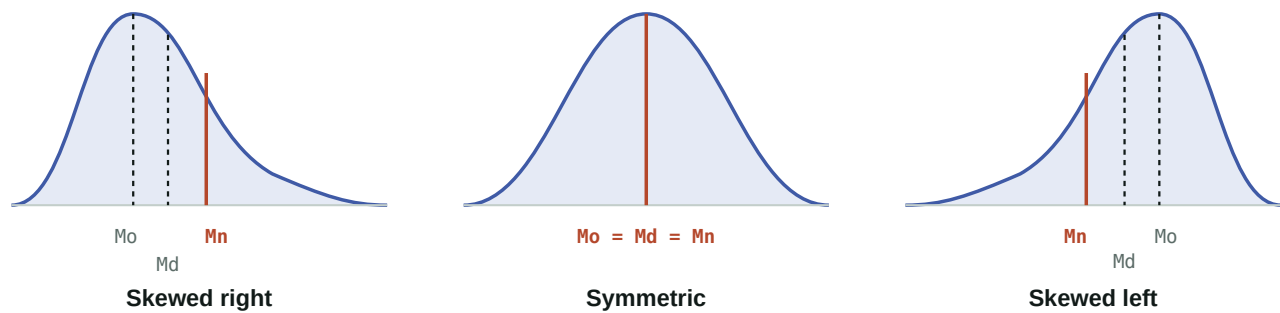
Describing a variable

Lecture 4 · Sep 8 · Pollock Ch. 3

Before you explain anything you describe it. Three questions about any variable: where is the middle, how spread out is it, and is it lopsided?

Mode	Most common value. The only average that works for nominal variables.
Median	The middle case when you line them up. Ignores extremes — use it when the data are skewed.
Mean	The arithmetic average. Gets dragged toward outliers.
Range	Highest minus lowest. Crude, and hostage to one weird case.
Standard deviation	The typical distance of a case from the mean. Small SD = cases bunched together.
Skew	A tail on one side. The mean always sits <i>toward</i> the tail.

FIGURE 4 · SKEW, AND WHY THE AVERAGE MOVES



Mo = mode, Md = median, Mn = mean. Income is the classic right-skewed variable: a handful of very rich people pull the mean far above the median, which is why "average income" flatters and "median income" describes. When a question asks which measure of central tendency to report, the answer is driven by skew.

THE ECOLOGICAL FALLACY – WORTH POINTS ON EVERY EXAM

Inferring something about **individuals** from data about **groups**. Poor counties voted for Bush, therefore poor people voted for Bush — that does not follow. The unit of analysis was the county, not the person. This is exactly the error in Frank's red-state map and in Messerli's chocolate study, which correlates *national* chocolate consumption with *national* Nobel counts and never observes a single laureate eating anything.

UNIT 4 Hypotheses and variables

Lecture 6 · Sep 15 · Pollock Ch. 4–5 · Quiz #1

Independent variable (X)	The proposed cause. The thing that does the explaining.
Dependent variable (Y)	The effect. The thing you're trying to explain — it "depends" on X.
Null hypothesis	There is <i>no</i> relationship between X and Y. This is what you actually test; you can only ever reject it or fail to reject it.
Rival hypothesis	A different explanation for the same pattern, usually built on a confounding variable.
Antecedent variable	A variable that comes <i>before</i> X and causes both X and Y — the source of spuriousness.
Intervening variable	A variable that sits <i>between</i> X and Y and transmits the effect. It explains the mechanism; it doesn't kill the claim.

THE TEMPLATE FOR A GOOD HYPOTHESIS

"As X increases, Y increases / decreases, across **units of analysis**."

Example: *As the perceived chance that neighbors will learn whether you voted increases, the probability of voting increases, across households.* Two variables, a direction, a unit, and a way to lose.

FOUR WAYS STUDENTS LOSE THESE POINTS

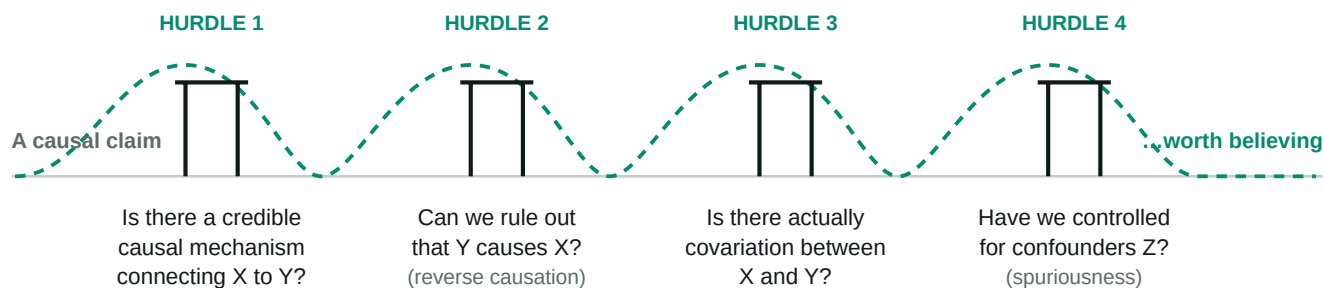
No variation: "Why do people vote?" studies only voters. **A definition, not a claim:** "Partisans are loyal to their party." **One case:** "Trump won because of the economy" — nothing to compare it to. **No direction:** "Education is related to turnout" doesn't say which way.

Causality — the four hurdles

Lecture 8 · Sep 22 · Kellstedt & Whitten Ch. 3 · Bueno de Mesquita & Fowler Ch. 3

This is the single most testable thing in the course. Kellstedt and Whitten give you four questions to run any causal claim through. Answer them yes / no / maybe. One confident *no* kills the claim. You never prove causation — you accumulate confidence.

FIGURE 5 · THE FOUR HURDLES



Knock over any one hurdle and the claim is done. Most public arguments clear only Hurdle 3 and stop there.

Hurdle 3 is the easy one, and it is the only one most people check. "Chocolate and Nobel prizes are correlated" clears Hurdle 3 and fails the other three.

Random assignment is powerful precisely because it clears Hurdles 2 and 4 at once: the researcher sets X, so Y can't have caused it, and randomization balances every confounder — including the ones nobody thought of.

THE THREE CASES KELLSTEDT & WHITTEN RUN THROUGH THE HURDLES

Life satisfaction and democratic stability — a showcase for Hurdle 2: do happy citizens produce stable democracies, or do stable democracies produce happy citizens? **Race and political participation. Head Start** — a Hurdle 4 problem, because families *select into* the program, so participants and non-participants differ before it begins.

FOUR THINGS PEOPLE GET WRONG ABOUT CAUSES

There is no such thing as *the* cause — outcomes have many, and every causal claim is relative to a stated counterfactual. A single counterexample doesn't refute a causal claim; causation here is about averages, not certainties. Causes precede effects. And a mechanism is not strictly required for an effect to be real.

What "cause" actually means

Bueno de Mesquita and Fowler define it with a **counterfactual**: X caused Y for a case if Y would have been different had X been different, with everything else held fixed. That definition creates the central problem of the whole field.

Potential outcomes

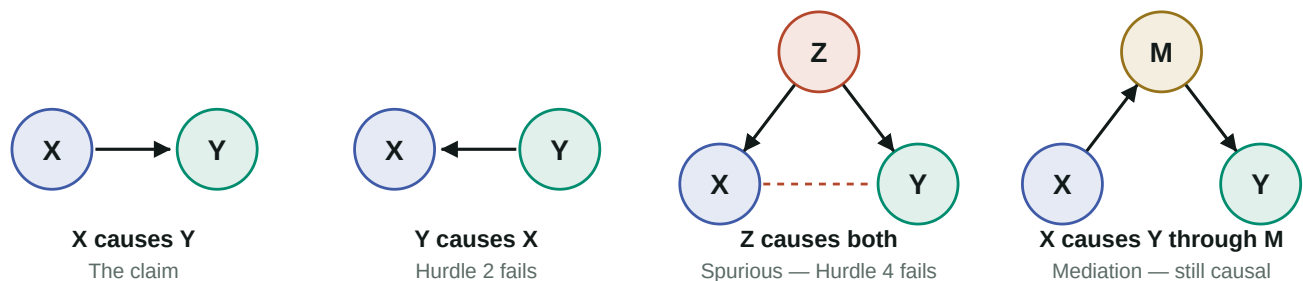
For each case there are two: Y(1), what happens if treated, and Y(0), what happens if not. The **treatment effect** is $Y(1) - Y(0)$.

Fundamental problem of causal inference

You only ever observe *one* of the two. The counterfactual is never seen. So a causal effect is never measured directly — it is inferred by finding a credible stand-in for the missing counterfactual. That stand-in is your control group.

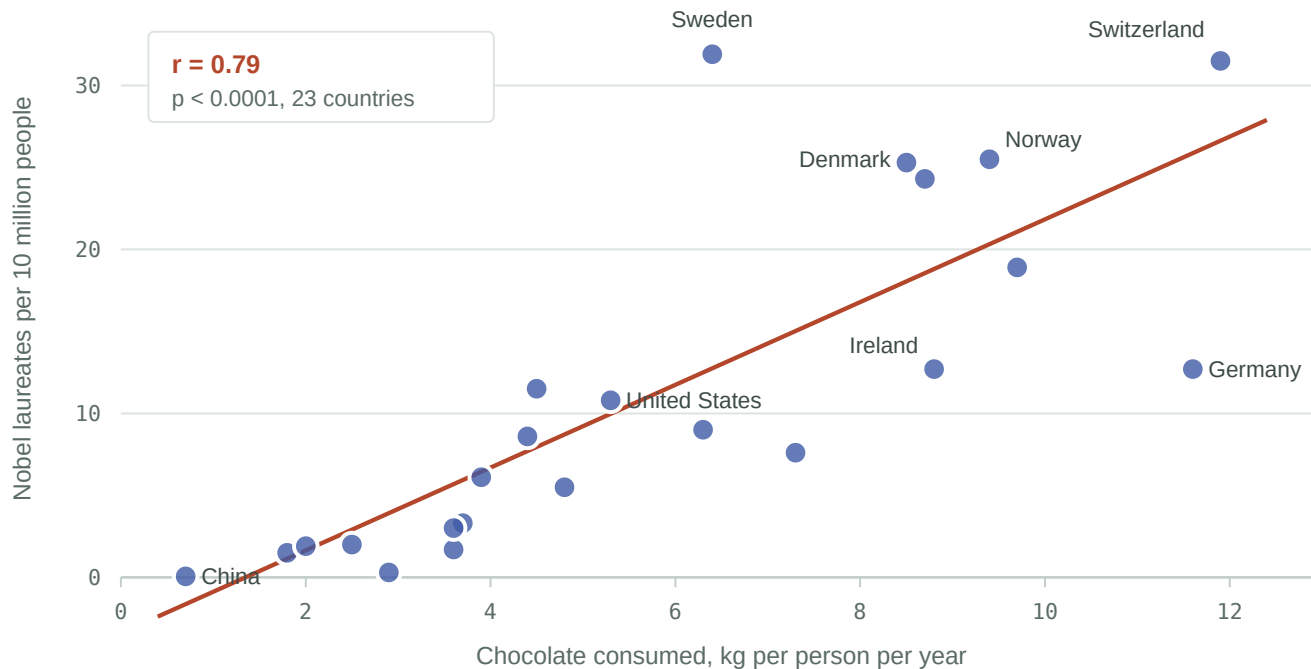
Average treatment effect (ATE)	The average of $Y(1) - Y(0)$ across cases. Effects can differ from person to person, so an ATE can be positive even where it's negative for some.
Causation without correlation	The one people forget: a real causal effect can show <i>zero</i> correlation if it runs in opposite directions for different groups, or if people compensate for it.

FIGURE 6 · FOUR THINGS A CORRELATION BETWEEN X AND Y CAN MEAN



Learn to tell the third panel from the fourth. An **antecedent** variable Z sits before X and destroys the claim. An **intervening** variable M sits between X and Y and explains it. Same-looking diagram, opposite verdict — a favorite exam distinction.

FIGURE 7 · THE MOST FAMOUS FAKE CAUSAL CLAIM IN THE SYLLABUS



Messerli (2012), *New England Journal of Medicine* — reconstructed from the published figure. A real, strong, highly significant correlation. Run it through the hurdles: no plausible mechanism at the national level, reverse causation is live (rich clever countries buy fancy chocolate), and every confounder you can name — wealth, education, research funding, being in Northern Europe — is uncontrolled. The unit of analysis is the country, so it also commits the ecological fallacy: not one laureate's diet was ever observed. Messerli wrote it as a joke; it is on the syllabus because the joke keeps getting published seriously.

UNIT 6

Research design and validity

Lecture 10 · Sep 29 · Johnson, Reynolds & Mycoff Ch. 6

A research design is the plan for which comparisons you'll make. The whole art is arranging comparisons so that only X differs between the groups.

INTERNAL VALIDITY

Is the relationship I found *inside this study* really causal? Experiments are strong here.

EXTERNAL VALIDITY

Does it generalize to other people, places and times? Lab experiments are often weak here.

THE DISTINCTION THAT SHOWS UP EVERY SINGLE SEMESTER

Random assignment ≠ random sampling. Random *assignment* puts subjects into treatment and control by chance — it buys **internal validity**. Random *sampling* draws subjects from a population by chance — it buys **external validity**. A study can have one without the other, and most have exactly one.

Threats to internal validity

THREAT	WHAT GOES WRONG
History	Something else happens between pretest and posttest — a scandal, a recession — and gets credited to your treatment.
Maturation	Subjects change on their own over time. Teenagers get more political with age whether or not you do anything.
Testing	Taking the pretest itself changes the posttest. Asking about voting makes people vote.
Instrumentation	The measuring device or the coder drifts. A survey question is reworded halfway through.
Regression to the mean	Extreme cases drift back toward average on their own. Target the worst schools and they'll improve regardless.
Selection bias	The groups differed <i>before</i> the treatment. The single biggest threat in observational work.
Attrition (mortality)	People drop out non-randomly, so the survivors aren't comparable anymore.
Demand effects / Hawthorne	Subjects behave differently because they know they're being watched.
Diffusion	The control group gets the treatment anyway — they talk to the treated group.

THE NAMED DESIGNS — KNOW THESE BY NAME

Classical randomized design: pretest and posttest, both groups. **Posttest-only control group design:** drops the pretest specifically to avoid testing effects. **Solomon four-group design:** combines both so the testing effect can be isolated. **Factorial design:** crosses multiple treatments. **Non-equivalent control group** and **interrupted time-series** are the two standard quasi-experiments.

And note the two ways to get control: **by design**, through randomization or matching, or **by statistical adjustment**, holding Z constant in the analysis. They are not equivalent — Unit 8 is about why.

The design menu, weakest to strongest on causality

DESIGN	WHO ASSIGNS X	INTERNAL	EXTERNAL
Case study / small-N	Nobody	Weak	Weak
Cross-sectional survey	Nobody	Weak	Strong if well sampled
Panel / longitudinal	Nobody	Better — time order is visible	Good
Quasi-experiment / natural experiment	Nature, a policy, a lottery	Good	Good
Field experiment	The researcher, in the real world	Strong	Strong
Lab experiment	The researcher, in a controlled setting	Strongest	Often weak

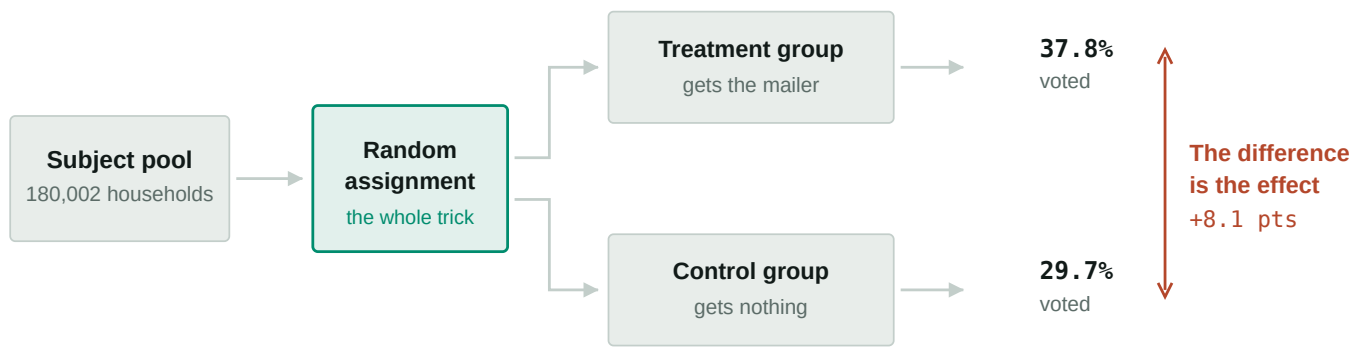
UNIT 7

Experiments and the logic of control

Lecture 12 · Oct 6 · Pollock Ch. 6 · Gerber, Green & Larimer · Bertrand & Mullainathan · Quiz #3

Randomization is the only technique that handles confounders **you never thought of**. Statistical controls can only adjust for variables you measured; random assignment balances everything at once, on average — observed and unobserved.

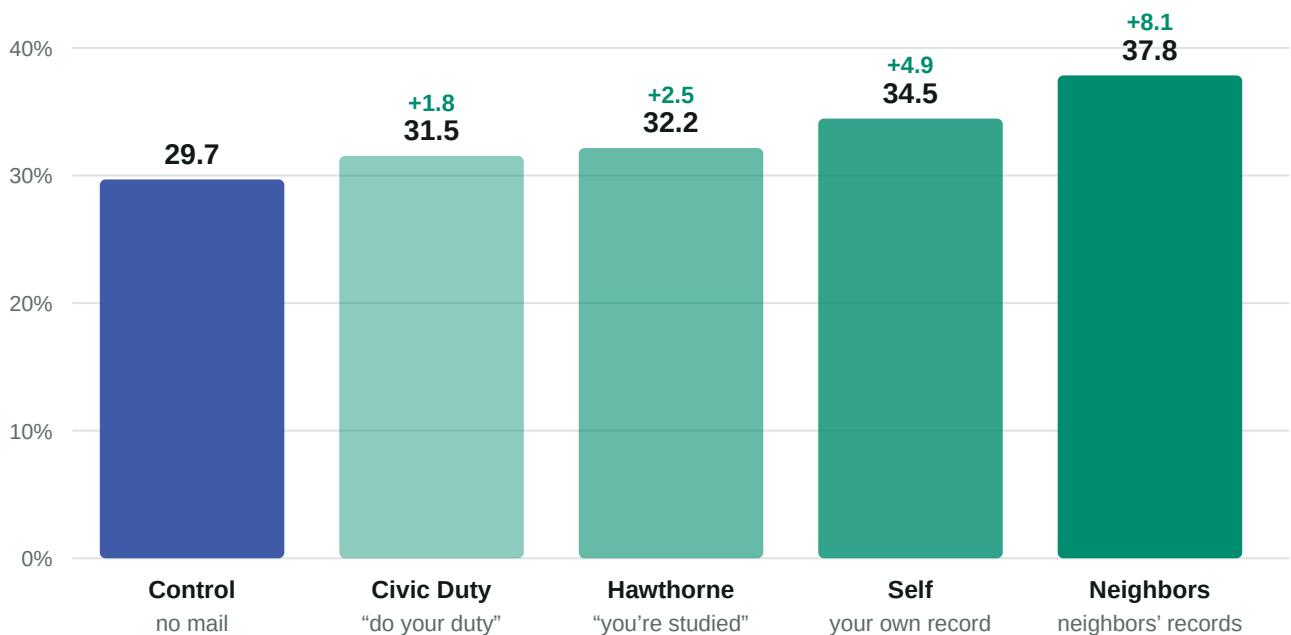
FIGURE 8 · ANATOMY OF AN EXPERIMENT



Because assignment was random, the two groups were alike in every respect before the mailer — so nothing but the mailer can explain the gap.

The check that randomization worked is the balance table: compare the groups on background characteristics before the treatment. Gerber, Green & Larimer report one, and the groups are statistically indistinguishable. If a study skips the balance table, ask why.

FIGURE 9 · GERBER, GREEN & LARIMER (2008) — SOCIAL PRESSURE AND TURNOUT



This is a dosage design, and that's the point. Four escalating treatments isolate *which* mechanism moves people. Priming civic duty barely works (+1.8). Being observed works more (+2.5). Showing your own record works more still (+4.9). Threatening to show the neighbors nearly doubles that (+8.1). Turnout isn't driven by information — it's driven by being seen. Context: single mailings before this typically moved turnout **less than one point**, and the 8.1-point effect rivals face-to-face canvassing at **\$1.93 per vote** for Neighbors and **\$3.24** for Self, against roughly \$20 per vote door-to-door. This experiment changed how American campaigns are run.

THE ETHICAL OBJECTION — BE ABLE TO STATE IT

The treatment worked by **shaming people**, using public records, without their consent, and it generated hundreds of angry phone calls to the researchers. Whether the knowledge gained justifies that is a live question. A good exam answer names the effect *and* the cost.

FIGURE 10 · BERTRAND & MULLAINATHAN (2004) — CALLBACK RATES BY NAME AND RÉSUMÉ QUALITY



~5,000 résumés, 1,300+ job ads in sales, admin, clerical and customer service, Boston and Chicago, 2001–02. Names were drawn from Massachusetts birth-certificate data to sound distinctively white or distinctively African-American. Overall: **9.65%** callback for white-sounding names vs. **6.45%** for Black-sounding names — a 3.2-point gap, **50% higher**, equivalent to about **eight extra years of experience**. The design solves a problem no survey can: employers see more about applicants than researchers do, so any observed gap could be something unmeasured. Randomly assigning the *name* to an otherwise identical résumé removes that possibility entirely. And notice the interaction — improving the résumé raised white callbacks by 2.29 points but Black callbacks by only 0.51 (not significant). The gap *widens* with qualifications.

HOW TO ANSWER "WHY IS THIS AN EXPERIMENT?" ON THE EXAM

Name three things: (1) the researcher **manipulated** the independent variable rather than observing it; (2) assignment to conditions was **random**, so treatment and control were equivalent beforehand on everything, measured or not; (3) the outcome was compared across groups, and the **difference** is the causal effect. Then say which hurdles it clears — 2 and 4 automatically, 3 by measurement, and 1 you still have to argue.

UNIT 8

Controlling without an experiment

Lectures 13–14 · Oct 8 & 13 · Pollock Ch. 7 · Leighley & Nagler Ch. 2

Most political questions can't be randomized — you can't assign someone a race, a region, or an income. The fallback is **controlled comparison**: split the sample by the suspected confounder *Z* and look at the *X–Y* relationship *inside* each slice. If it survives, *X* looks real. If it vanishes, it was spurious.

WHAT YOU SEE AFTER CONTROLLING FOR <i>Z</i>	WHAT IT MEANS	VERDICT ON <i>X</i>
The relationship holds inside every slice	<i>Z</i> wasn't doing the work	X survives
The relationship disappears in every slice	<i>Z</i> caused both <i>X</i> and <i>Y</i>	Spurious
The relationship is strong in one slice, absent in another	Interaction — <i>X</i> matters, but only under certain conditions	Conditional

Bartels' Kansas analysis is exactly this. Uncontrolled, white non-college voters moved 5.9 points Republican over fifty-two years. Control for region and the effect is entirely Southern:

SLICE	CHANGE, 1952–2004	READING
All white non-college voters	–5.9	Looks like Frank is right
...in the South	–19.7	Huge — but the South realigned over race, not culture-war economics
...outside the South	–1.0	Fifty-two years, one point. Nothing happened.

THE PERMANENT LIMIT OF STATISTICAL CONTROL

You can only control for variables you **measured and thought of**. Randomization balances the ones you didn't. That asymmetry is why an experiment on a small, artificial question often beats a giant observational study on the real one — and it's the standard critique to deploy on any exam question about survey-based causal claims.

Leighley & Nagler: two different questions, two different methods

This chapter is on the syllabus because it shows when to control and when *not* to. The two questions look identical and aren't:

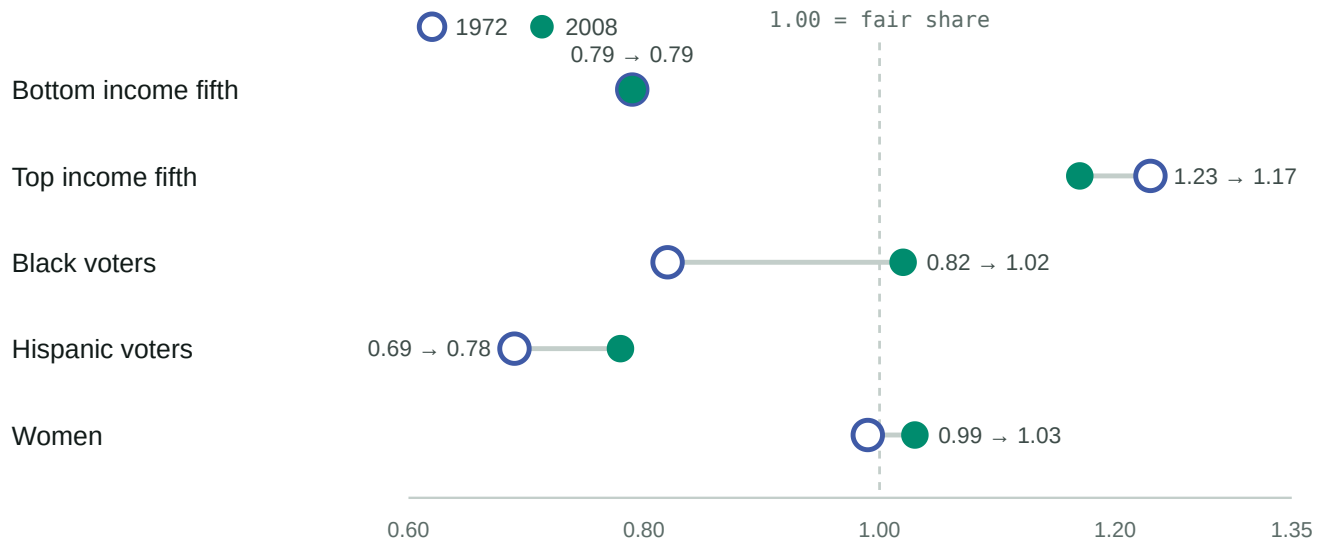
1. **"Are poor people underrepresented among voters?"** — Don't control for anything. Just compare shares. Controlling for age or education here would answer a different question.
2. **"Is a poor person less likely to vote than an otherwise identical richer person?"** — You must control, because income rises with age and young people vote less.

Their tool for the first question is the **representativeness ratio**: a group's share of actual voters divided by its share of the citizen voting-age population. **1.0 = proportional. Above 1 = overrepresented. Below 1 = underrepresented.**

RELATIVE MEASURES BEAT ABSOLUTE ONES ACROSS DECADES

They use income *quintiles* and education *thirds*, not dollar brackets and degree categories, because the meaning of a category drifts. College graduates went from **12.0%** of adults in 1972 to **29.4%** in 2008, while adults without a high-school diploma fell from **41.8%** to **13.4%**. Measured in absolute categories, the least-educated group looks like its representation collapsed — 0.79 down to **0.62**. Measured as the bottom third, it barely moved: 0.79 to **0.78**. Same data, two opposite stories, purely from the choice of measure.

FIGURE 11 · WHO IS OVERREPRESENTED AMONG VOTERS, 1972 VS. 2008



The punchline is the top line: it didn't move. Across thirty-six years of exploding economic inequality, the bottom income fifth's representation among voters was **0.79 in 1972 and 0.79 in 2008**. Race is where the change happened — Black voters went from underrepresented to slight parity. Two more findings worth memorizing: turnout has *not* declined (58.4%–65.5% throughout, once you use citizens as the denominator), and the apparent decline everyone cites is a **denominator artifact** — noncitizens grew from under 2% of the voting-age population to 8.4%, so counting them makes turnout look like it's falling when it isn't. Hispanic figure is 1976, their first year of data.

TWO MORE FINDINGS FROM THIS CHAPTER

Age moved a lot. Voters aged 18–24 are the lowest-turnout group in every single year, with a ratio around 0.77 throughout. Meanwhile the 76–84 group went from underrepresented at **0.91** to overrepresented at **1.11**, and their share of all votes cast rose from 3.6% to 6.0%.

Counterfactual arithmetic. Had Black turnout in 2008 stayed at its 1972 level, Obama's vote share would have been about **1.2 points lower**.

THE CAVEAT THEY RAISE ON THEMSELVES

Measured against the *total* voting-age population rather than citizens, income bias **has** increased — the 2008 bottom-quintile ratio is 0.79 on citizens but **0.75** on the full voting-age population. As they put it: do you care about representation of the *eligible*, or of the *available*? Same data, different denominator, different answer. Again.

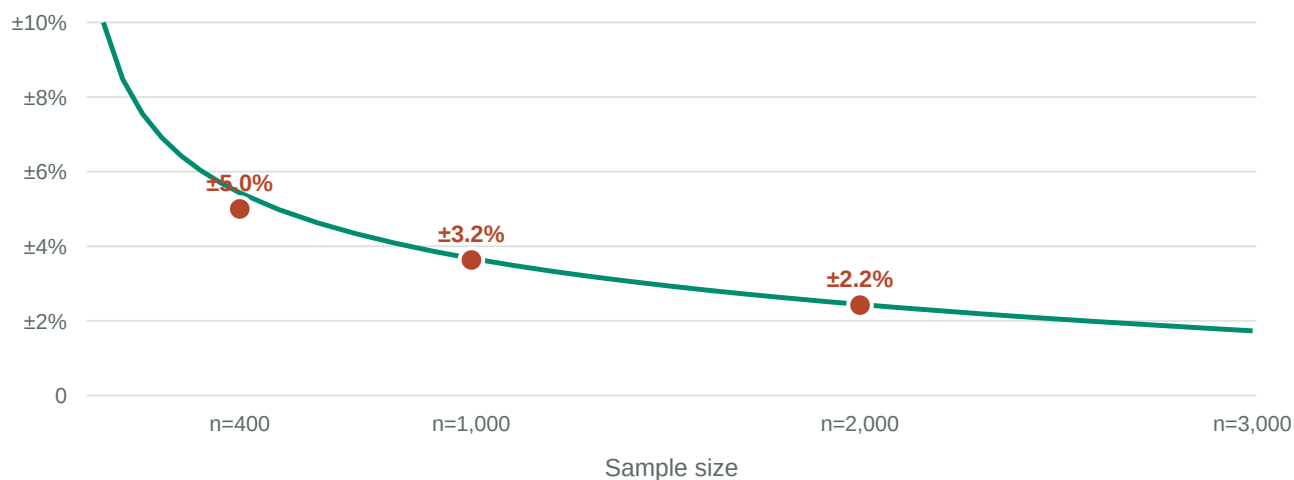
UNIT 9 Sampling

Lecture 18 · Oct 27 · Pollock Ch. 8 (p. 193–96) · Moore & Notz p. 35–40, 57–71

Population	Everyone you want to describe.
Sampling frame	The list you actually draw from. The gap between frame and population is where coverage error lives.
Parameter vs. statistic	Parameter = the true number in the population (fixed, unknown). Statistic = the number in your sample (known, varies sample to sample). Statistics estimate parameters.
Simple random sample	Chosen so that <i>every set of n individuals</i> has an equal chance of being the sample. Note the wording — "every individual has an equal chance" is necessary but not sufficient.

Stratified sample	Divide the population into groups, draw randomly within each. Guarantees representation of small groups.
Convenience sample	Whoever is easy to reach. Systematically unrepresentative.
Voluntary response sample	People select themselves in — call-in polls, opt-in web surveys. Overrepresents strong, usually negative, opinions.
Confidence statement	Always has two parts: a margin of error and a confidence level. Reporting one without the other is incomplete.
Bias vs. variability	Bias = consistently off target; fixed only by <i>better design</i> . Variability = scattered; fixed by a <i>bigger sample</i> . Same dartboard as Figure 2.

FIGURE 12 · MARGIN OF ERROR SHRINKS WITH THE SQUARE ROOT OF N



The quick rule: margin of error $\approx 1 \div \sqrt{n}$. A thousand people gets you about ± 3 points. To halve that you need *four times* the sample — 4,000 people for ± 1.6 . This is why almost every national poll is 800–1,500 people: past that, you pay a lot for very little. Two more things that trip students up: the **population size doesn't matter** (1,500 is as good for Nashville as for the United States, so long as the population is at least $\sim 100\times$ the sample), and this curve covers *only* random sampling error — none of the errors in the next unit.

LITERARY DIGEST, 1936 – THE EXAMPLE THAT WILL BE ON A QUIZ

The *Literary Digest* mailed **10 million** ballots and got **2.4 million** back, then confidently predicted Landon would beat Roosevelt. Roosevelt won in a landslide. Their sample was drawn from car registrations, phone books and magazine subscribers — affluent in 1936 (**coverage error**) — and respondents chose to mail it in (**voluntary response**). Gallup used a far smaller but better-designed sample and got it right. **A huge sample cannot fix a biased design.**

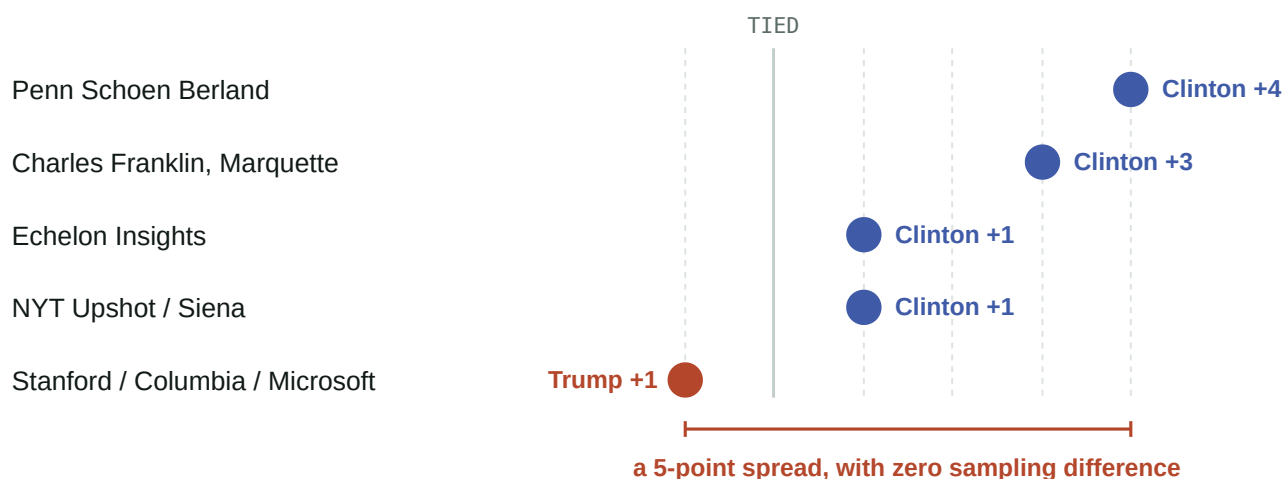
UNIT 10 Why polls miss

Lecture 19 · Oct 29 · Moore & Notz p. 41–48 · Cohn · Clinton · Hillygus · Quiz #4

The single idea of this unit: the reported margin of error is the smallest error in a poll. It covers random sampling only. Everything below is bigger, and none of it shrinks when you interview more people.

ERROR TYPE	WHAT HAPPENS	FIXED BY A BIGGER SAMPLE?
Sampling error	Random luck of the draw. The only thing in the $\pm 3\%$.	Yes
Coverage error	Groups missing from the frame entirely — cell-only households, non-English speakers.	No
Nonresponse	Selected people can't be reached or refuse. The dominant problem today.	No
Measurement / response bias	People misreport — social desirability, memory, interviewer effects. Turnout overreporting is the classic.	No
Question wording & order	Leading or loaded wording changes answers; so does what you asked first.	No
Adjustment error (weighting)	The pollster's judgment calls about what the electorate looks like.	No
Likely-voter modeling	Deciding who actually shows up. Hillygus: often the most consequential choice of all.	No

FIGURE 13 · SAME 867 INTERVIEWS, FOUR POLLSTERS, FOUR ANSWERS



Nate Cohn's natural experiment (NYT Upshot, 2016). Four outside pollsters got the identical raw interviews from a Florida poll. The differences come entirely from two judgment calls: **how to weight** the sample and **who counts as a likely voter**. Their implied electorates differed too — white share 65–70%, partisan balance from D+5 to R+1. Cohn's conclusion is quotable: the reported margin of error "doesn't even come close to capturing total survey error." Joshua Clinton's 2024 piece makes the same point differently — sixteen defensible weighting schemes on one survey produced margins from **Harris +3.8 to Harris +9.4**.

HILLYGUS: POLLING ERROR IS NOT A MODERN CRISIS

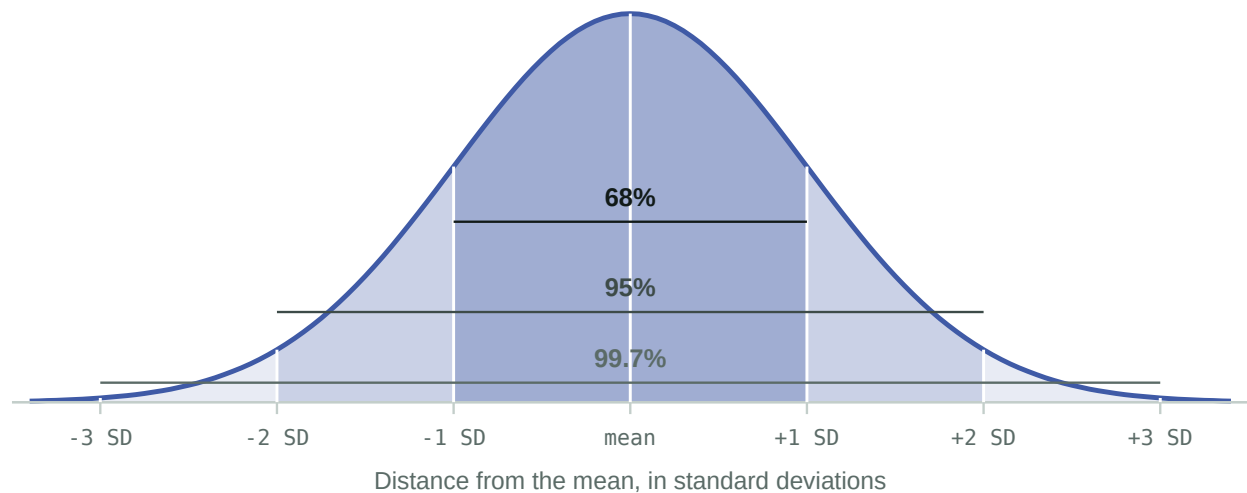
She catalogs the error types — coverage, nonresponse, measurement, processing, adjustment — and argues the **likely-voter screen** is often the most consequential choice a pollster makes. Her showcase: Gallup in **2000**, where changes to the likely-voter model produced a **19-point swing** in the reported margin over a matter of days, none of it real movement. Add the two canonical failures: **1948**, Dewey defeats Truman, where pollsters simply stopped polling weeks out; and **New Hampshire 2008**, where every poll had Obama winning and Clinton won.

WHAT TO DO WITH POLLS, PER THESE READINGS

Average many polls rather than trusting one — but know the limit: averaging cancels random noise, not **house effects**. If every pollster misses cell-only households the same way, the average is just as wrong. Also: only about **5 of every 100** contacted registered voters respond to a survey at all, so weighting isn't a technical detail bolted on at the end — it is doing most of the work.

You need the normal curve for one reason: it describes how **sample statistics** behave. Draw sample after sample from the same population and the sample means pile up in a bell shape centered on the truth. That fact is what lets you say anything about a population from one sample.

FIGURE 14 · THE 68–95–99.7 RULE



Learn these three numbers cold. About **68%** of cases fall within one standard deviation of the mean, **95%** within two, **99.7%** within three. The **z-score** converts any value into this scale: $z = (\text{value} - \text{mean}) \div \text{standard deviation}$. A z of 2 means "two SDs above average — only about 2.5% of cases are higher."

Sampling distribution

Not the distribution of your data — the distribution of the *statistic* across all the samples you could have drawn. It is normal, and centered on the true parameter.

Standard error

The standard deviation of that sampling distribution. It measures how much your estimate would bounce around from sample to sample. Bigger $n \rightarrow$ smaller standard error.

Central limit theorem

Sample means are normally distributed even when the underlying variable isn't, as long as n is decently large. This is why the normal curve is everywhere.

You have a sample. You want to say something about a population you can't see. Inference is the bridge, and it comes in two forms: an **interval** around your estimate, and a **test** of whether a relationship is real.

FIGURE 15 · WHAT “95% CONFIDENCE” ACTUALLY MEANS



19 of 20 intervals capture the truth. That is the 95%. It is a property of the method, not of any one interval.

The interpretation students get wrong: a 95% confidence interval does *not* mean “there’s a 95% chance the true value is in this interval.” The true value is fixed; it’s either in there or it isn’t. What’s 95% is the *procedure* — do this over and over and 19 times in 20 you’ll capture it. Formula worth memorizing: **estimate $\pm$ (1.96 $\times$ standard error)**.

Hypothesis testing in four steps

1. State the **null hypothesis**: there is no relationship between X and Y.
2. Assume the null is true.
3. Ask: *if the null were true, how likely is a relationship at least this big, just from sampling luck?* That probability is the **p-value**.
4. If $p < .05$, reject the null — the result is “statistically significant.” Otherwise you **fail to reject** it. You never “accept” the null.

	IN REALITY: NO RELATIONSHIP	IN REALITY: A RELATIONSHIP
You say: relationship	Type I error — a false positive. You announced something that isn’t there. $p < .05$ caps this risk at 5%.	Correct
You say: no relationship	Correct	Type II error — a false negative. Usually caused by too small a sample.

STATISTICALLY SIGNIFICANT $\neq$ IMPORTANT. TWO DIFFERENT MISTAKES.

Big samples make trivial effects significant. With 300,000 cases, a difference of a tenth of a point clears $p < .05$ and means nothing.

Small samples hide real effects. Significance answers “is this probably not zero?” — not “is this big enough to care about?” Levitt’s capital-punishment estimate is the model case: even accepting the most generous deterrence numbers, executions explain about **1.5%** of the 1990s homicide decline. Real, arguably significant, and far too small to be the answer.

And the one that costs the most points: **significance says nothing about causality.** A significant correlation still has to clear all four hurdles.

UNIT 13 Correlation and regression

Lecture 28 · Dec 1 · Pollock Ch. 11

Correlation puts a number on how tightly two variables move together. Regression goes further and gives you a **prediction rule** — and a sentence you will be asked to write on the final.

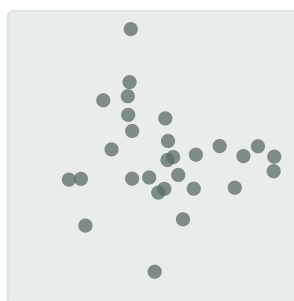
FIGURE 16 · READING A CORRELATION COEFFICIENT



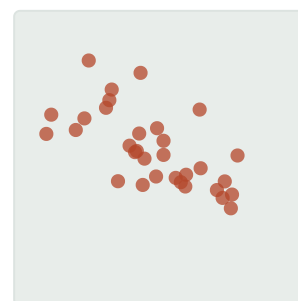
$r = +0.92$
strong, positive



$r = +0.45$
moderate, positive



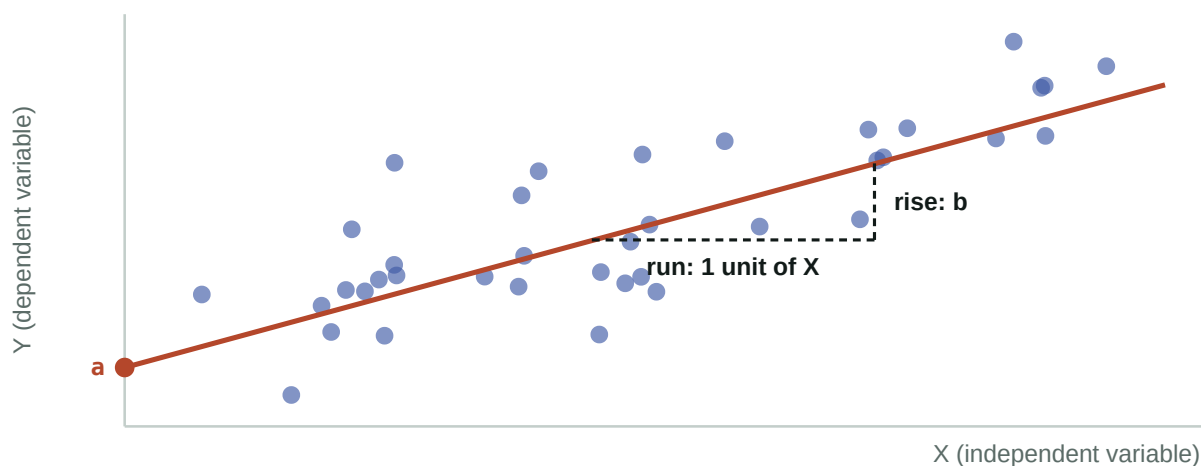
$r = 0.00$
no linear relationship



$r = -0.75$
strong, negative

Sign = direction, size = strength. Pearson's r runs from -1 to $+1$. Two cautions: $r = 0$ means no *straight-line* relationship — a perfect U-shape has r near zero. And r^2 (r squared) is the share of the variation in Y that X accounts for; $r = 0.79$ for chocolate and Nobels means $r^2 \approx 0.62$, which sounds impressive and still isn't causal.

FIGURE 17 · WHAT THE REGRESSION SLOPE MEANS



$$Y = a + bX$$

a = the intercept, Y when X is zero · b = the slope

Write this sentence on the exam: "A one-unit increase in X is associated with a change of b units in Y ." Note "associated with," not "causes" — the slope is a description of the data, and causality still has to clear the four hurdles. In **multiple regression**, each b is the effect of that variable *holding all the others constant*. That is statistical control, and it is why regression tables have so many rows.

HOW TO READ A REGRESSION TABLE

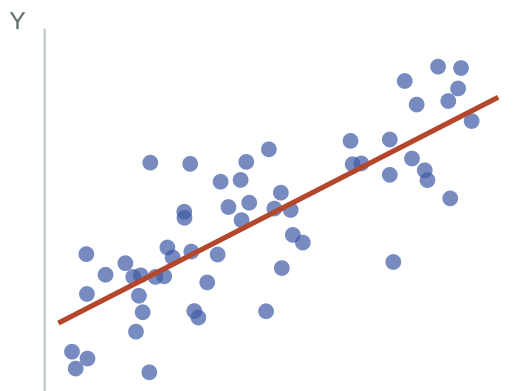
Each row is a variable. The first number is the **coefficient (b)** — direction and size of the effect. Beneath or beside it, in parentheses, is the **standard error** — how precise that estimate is. Asterisks or a p -column tell you significance. **Rule of thumb:** if the coefficient is more than about twice its standard error, it is significant at $p < .05$.

When you only have a handful of cases you have to choose them deliberately. King, Keohane and Verba give one rule that decides most of it:

THE RULE

Select cases on the independent variable. Never on the dependent variable. Never on both.

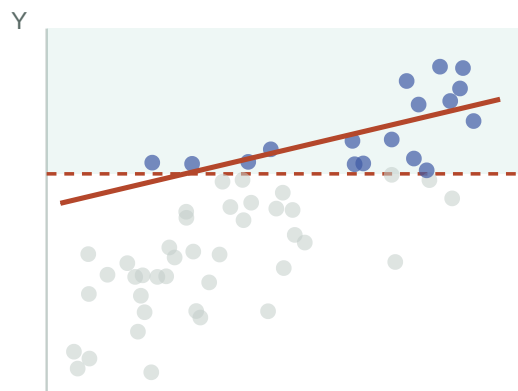
FIGURE 18 · WHAT SELECTING ON THE DEPENDENT VARIABLE DOES TO YOUR ANSWER



All cases

every value of Y is in the sample

estimated slope $b = 0.68$



Only high-Y cases

"let's study the successes"

estimated slope $b = 0.32$

Same data, same true relationship, half the estimated effect. Keeping only high-Y cases **biases the estimate toward zero** — it makes X look weaker than it is. The one consolation: because the direction of the bias is known, a significant effect found in a truncated sample means the true effect is *at least* that large. Selecting on *both* X and Y is worse — the bias can run in any direction and you cannot know which.

- **No variation in Y = zero information.** Study only revolutions and you cannot learn what causes revolutions — you never see the countries that had the same conditions and didn't revolt. Same for studying only successful startups, only school shooters, only wars.
- **Selecting on X is fine.** Deliberately picking cases with very different values of X actually *increases* your leverage. The cost is only that your findings apply to that range of X.
- **Any selection rule correlated with Y creates the bias** — even one you never thought of as selecting on Y. Using only cases covered in newspapers, only "important" cases, only cases with available data, only organizations that survived long enough to study.
- **Random selection is risky with small N.** With six cases a random draw can easily be unrepresentative. That is why KKV recommend *intentional* selection on X for qualitative work.

THE FREAKONOMICS PAIRING

Venkatesh got a crack gang's actual financial records — an extraordinary data source on a population nobody can survey, showing foot soldiers earning below minimum wage while the leader took a large share. What one gang's books **can** do is show a mechanism and kill a stereotype. What they **cannot** do is establish a general causal claim: one non-randomly obtained case, no comparison group, no variation on the outcome. Read it for what a single case is good for and what it isn't.

The audio editions

Two narrated versions of this guide, delivered as MP3s in the conversation. Every figure on this page is *described in words* in the recordings, so you don't need to be looking at anything. Timestamps below — the highlighted rows are the highest-value stretches.

Condensed

43:48

The whole course, tightened for repeat listening. Every key concept and every number that matters. This is the one to play on the walk to class.

0:00	Welcome
0:44	The one idea
1:45	Theory and hypotheses
3:48	Where numbers come from
5:20	Measurement
7:30	Why measurement decides Kansas
9:41	Describing data
10:57	The four hurdles
12:56	What cause means
14:00	Reading causal diagrams
15:03	Validity and research design
16:39	Experiments
20:33	Controlling without an experiment
23:23	Sampling
25:48	Why polls miss
28:04	Probability and inference
31:22	Correlation and regression
33:01	Small numbers and selection bias
35:22	The seven debates
41:28	The exam
43:20	Close

Full

84:54

All fourteen units at full depth, the seven debates, and a **spoken self-test** — twelve questions, each followed by seven seconds of silence, then the answer. Use it for a deep pass before the midterm and again before the final.

0:00	Welcome
0:46	The shape of the course
2:13	Unit 1, Theory driven research
6:15	Unit 2, Concepts and measurement
9:45	The Kansas argument
12:37	Unit 3, Describing a variable
14:54	Unit 4, Hypotheses and variables
16:59	Unit 5, Causality, part one
19:38	Unit 5, Causality, part two
23:06	Reading causal diagrams
24:09	Unit 6, Research design
28:46	Unit 7, Experiments
34:41	Unit 8, Controlling without an experiment
40:36	Unit 9, Sampling
45:04	Unit 10, Why polls miss
49:19	Unit 11, Probability and the normal curve
51:24	Unit 12, Inference and significance
55:16	Unit 13, Correlation and regression
58:36	Unit 14, Small numbers and selection bias
62:21	The seven debates
70:22	Levitt's accounting
72:01	The five exam frames
74:39	Self test
81:02	Glossary
84:14	Close

For Quiz 1 on September 15, the stretch you want is **1:45 to 9:41** of the condensed recording — theory, hypotheses, where numbers come from, measurement, and the Kansas debate.

The seven debates

Each week Trounstein pairs a popular claim with a scholar who tests it. If you can fill in **Claim** → **Test** → **Verdict** for all seven, you can answer most of the reading questions on any quiz.

What's the matter with Kansas?

LECTURE 5 · SEP 10

CLAIM **Thomas Frank:** working-class whites are voting against their economic interests because the GOP mobilizes them on culture. Evidence: the poorest county in America went for Bush by over 75%.

TEST **Bartels:** 14 elections of ANES survey data, 1952–2004, with region and income controlled. **Ruffini:** the same question two decades later.

VERDICT Frank's thesis doesn't survive. The shift is –5.9 points overall, –19.7 in the South, and –1.0 outside it. By income, poor whites moved *toward* the Democrats (+4.5). Economic issues outweighed abortion in vote choice by more than 2 to 1. In 2004 abortion ranked **13th of 15 issues** for non-college whites.

Methods lesson: operationalization decides the answer; ecological fallacy; controlling for a lurking variable (region).

Who receives public assistance?

LECTURE 7 · SEP 17

CLAIM Americans oppose welfare because of how they picture the poor — and the picture is racialized.

TEST **Gilens:** content analysis. Code the racial composition of poor people shown in news magazine photographs and network TV coverage, then compare it to the actual composition of the poor in Census and CPS data.

VERDICT The media portrait is systematically unrepresentative of the real poverty population, and public misperception tracks the portrait rather than the data.

Methods lesson: content analysis as a measurement strategy; comparing a *measured perception* against a *measured reality*; where public opinion data comes from.

Is stop-and-frisk racial profiling?

LECTURE 9 · SEP 24

CLAIM **Bloomberg:** stop rates should be benchmarked against the descriptions of criminal suspects, not against the population — so the racial disparity isn't profiling.

TEST **Bertrand & Mullainathan:** stop arguing about the right denominator. Randomly assign race — via the name — to otherwise identical résumés and mail 5,000 of them out.

VERDICT 9.65% callback for white-sounding names vs. 6.45% for Black-sounding names. A 50% gap, worth about eight years of experience, and it *widens* as résumés improve.

Methods lesson: the whole fight in the op-ed is about choosing a comparison group. An experiment makes the comparison group by construction.

Does money buy votes?

LECTURE 11 · OCT 1

CLAIM Contributions correlate with roll-call votes, therefore donors buy legislators.

TEST **Wright:** the correlation is a weak design — donors give to legislators who *already* agree, so causation may run backwards. **Hertel-Fernandez et al.:** survey 101 senior Congressional staffers and test their beliefs about constituent opinion against MRP estimates from the CCES.

VERDICT Staffers misjudge their constituents badly and in a conservative direction — 91% underestimated support for gun background checks, 78% underestimated support for CO₂ limits. A list experiment found **45%** had changed their view after talking to a donor group. The channel isn't buying votes; it's shaping what legislators believe voters want.

Methods lesson: reverse causation (Hurdle 2); the list experiment as a way to ask an embarrassing question; the difference between an effect and its mechanism.

Who votes, and is it getting worse?

LECTURE 14 · OCT 13

- CLAIM** Turnout is falling and, as inequality grows, the electorate is getting richer relative to the country.
- TEST** **Leighley & Nagler:** Current Population Survey data, 1972–2008, using representativeness ratios and relative measures (income quintiles, education thirds) rather than absolute categories.
- VERDICT** Both claims fail. Turnout ran 58.4%–65.5% with no trend; the "decline" is an artifact of counting noncitizens in the denominator. And income bias was **identical in 1972 and 2008** (bottom-fifth ratio 0.79 both years).

Methods lesson: denominators change conclusions; relative vs. absolute measures over time; when to control and when not to.

Are independents rising?

LECTURE 20 · NOV 3

- CLAIM** **Kahn, Brooks:** a record 40% of Americans call themselves independent — the largest and most decisive bloc in politics.
- TEST** **Keith et al.:** push independents on which party they lean toward. **Fowler et al.:** model 280,000+ CCES respondents' actual positions across 20+ policy questions instead of trusting a self-applied label.
- VERDICT** Most "independents" are closet partisans who lean and then vote that way; Abramowitz puts truly uncommitted voters at **under 10%** of actual voters. Fowler et al. find roughly **73%** of Americans are well described by one left–right dimension, **21%** hold genuine but idiosyncratic views, and **6.5%** are just answering inattentively — and inattentive respondents give *more* extreme answers than actual extremists.

Methods lesson: a self-reported label is not the concept; measurement error can masquerade as a real category.

Why did crime fall in the 1990s?

LECTURE 25 · NOV 19

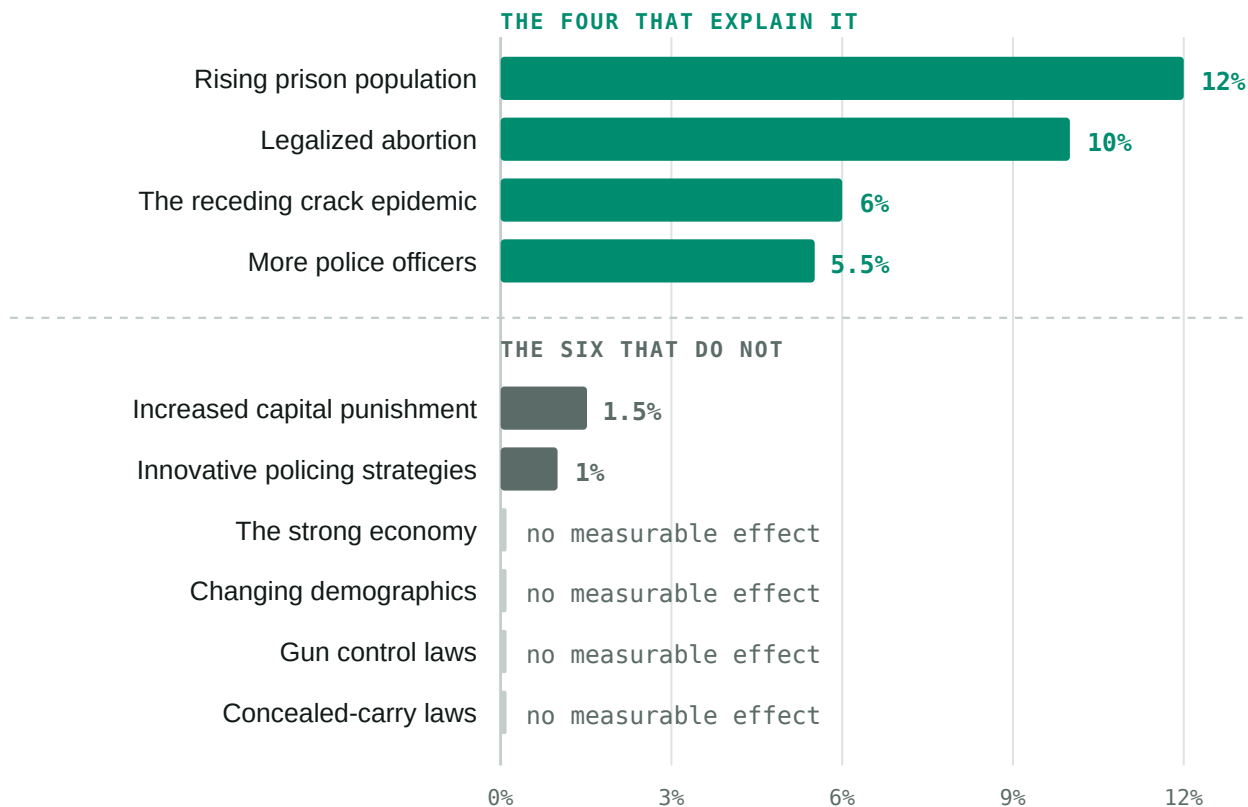
- CLAIM** **Gladwell:** crime is an epidemic with a tipping point; broken-windows policing tipped New York.
- TEST** **Levitt:** take the best-identified elasticity for each proposed cause and multiply by how much that factor actually changed. **Sampson & Winter:** follow a Chicago birth cohort for 18 years and triangulate OLS, matching and instrumental variables on childhood lead.
- VERDICT** Homicide fell **43%** nationally — everywhere, not just New York. San Diego fell 72.8%, Austin 69.5%, San Jose 69.2% against New York's 73.6%, and NYC hired police at three times the national rate. Sampson finds lead raises antisocial behavior but shows *no* direct effect on arrests, and warns that individual-level results don't settle the aggregate question.

Methods lesson: effect-size accounting; a mechanism story is not a test; the ecological fallacy in reverse; triangulating methods with different assumptions.

Worked example: Levitt's accounting of the crime decline

This is the best single illustration in the course of the difference between *an effect exists* and *an effect matters*. Levitt takes each proposed cause, uses the best available estimate of its effect, multiplies by how much that factor actually changed between 1991 and 2001, and sees what's left over.

FIGURE 19 • ESTIMATED SHARE OF THE 1990S HOMICIDE DECLINE EXPLAINED, BY FACTOR



Homicide fell 43% between 1991 and 2001; these four factors account for roughly 36 points of it. Now compare that to what the ten largest newspapers actually wrote about over the same decade: innovative policing was mentioned **52 times**, prisons 47, the strong economy 28 — and legalized abortion, Levitt's second-largest factor, was mentioned **zero times**. Popular explanations are ranked by how satisfying they are, not by how much they explain. Levitt's closing puzzle is worth remembering too: run the same accounting on 1973–1991 and crime should have fallen then. It didn't. "The real puzzle is not why crime fell in the 1990s, but why it did not start falling sooner."

Exam kit

Six quizzes, a midterm and a cumulative final. Each quiz has a take-home part posted at 9am the day before and due before class, plus an in-class part — so the take-home is where you build the notes you'll want later.

Sep 15 Quiz 1 — Units 1–3. Provenance of numbers, theory, measurement, the Kansas debate.	Sep 24 Quiz 2 — + hypotheses, Gilens, the four hurdles.	Oct 6 Quiz 3 — + research design, money and votes, experiments.	Oct 15 Midterm — everything through Oct 13. Units 1–8.
Oct 29 Quiz 4 — sampling, polling error, weighting.	Nov 10 Quiz 5 — independents, probability, the normal curve.	Dec 3 Quiz 6 — significance, crime, regression, campaigns.	Dec 17 Final , 9–11am. Cumulative — all fourteen units.

The five questions that keep coming back

Trounstine's stated goal is that you can "analyze, explain, and apply" to *new* contexts — so expect an unfamiliar claim and a request to evaluate it. These five frames cover most of it.

1 · "HERE IS A CLAIM FROM THE NEWS. EVALUATE IT."

Run the four hurdles in order and say yes / no / maybe to each with a reason. Name the specific confounder you're worried about, not "other factors." Finish with what evidence *would* settle it — usually "an experiment that randomly assigned X," and then explain why that's impossible here and what the next-best design would be.

2 · "DESIGN A STUDY TO ANSWER THIS QUESTION."

State the hypothesis with a direction. Name the unit of analysis. Give an operational definition for X and Y and defend it against a validity objection. Say how you'd assign or approximate assignment of X. Name your control group and what makes it a credible counterfactual. Then admit the biggest threat to internal validity that remains.

3 · "WHY IS THIS MEASURE BAD?"

Separate validity from reliability. Validity: does the operational definition capture the concept — Nobel prizes for cognitive function, "no college degree" for working class, self-reported turnout for turnout? Reliability: would repeating the procedure give the same answer? Then say which direction the error runs, because systematic error is far worse than noise.

4 · "THIS POLL SAYS X. HOW MUCH SHOULD YOU TRUST IT?"

Sampling error is the smallest error present. Ask about the frame (who's missing), response rate (how few answered), weighting (whose electorate is assumed), and the likely-voter screen. Cite Cohn — five estimates from identical data, spanning five points — or Clinton — sixteen weightings, spanning six points. Recommend averaging polls, and note that averaging doesn't fix house effects.

5 · "INTERPRET THIS TABLE / COEFFICIENT / CHART."

Say the direction, the size in real units, and whether it's significant (coefficient more than twice its standard error). Then say what it doesn't tell you. In a multiple regression, always add "holding the other variables constant." Never write "causes" unless the design earns it.

Quick self-test

Answer out loud, then open. If you can't do these cold, that's your study list.

? Name the four hurdles, in order.

1. Is there a credible causal mechanism connecting X to Y? **2.** Can we rule out that Y causes X? **3.** Is there covariation between X and Y? **4.** Have we controlled for all confounding variables Z? One confident "no" defeats the claim.

? What's the difference between random sampling and random assignment?

Random **sampling** picks who is in your study — it buys **external validity** (generalizability). Random **assignment** decides who gets the treatment — it buys **internal validity** (causal confidence). A national survey has the first and not the second; a lab experiment on 60 undergraduates has the second and not the first.

? Why can't you ever directly measure a causal effect?

The **fundamental problem of causal inference**: a causal effect is the difference between what happened and what *would have* happened, but for any case you only ever observe one of those. The counterfactual is never seen — so every method is a scheme for constructing a believable stand-in for it.

? A study finds counties with more churches have more crime. What's wrong?

Two things. **Spuriousness**: population size causes both — bigger counties have more of everything. And the **ecological fallacy**: even if the county correlation were real, it says nothing about whether churchgoers commit crimes. The unit of analysis is the county, not the person.

? A poll of 1,000 has a ± 3 point margin of error. What is that number covering?

Random sampling error only. It says nothing about coverage error (who's missing from the frame), nonresponse (only ~5 in 100 contacted people answer), response bias, question wording, weighting choices, or the likely-voter model. Those are usually larger, and none of them shrink when you interview more people.

? Why is "no college degree" a bad operational definition of "working class"?

It doesn't capture the concept — 40% of white non-college voters in 2004 had family incomes over \$60,000, and more than half called themselves middle class. It's a **validity** failure. Swap in income and Bartels' answer flips sign: the poorest white voters moved *toward* the Democrats.

? You want to know what causes civil wars, so you study twenty civil wars. What's the error?

Selecting on the dependent variable, with no variation in Y. Every case has the same outcome, so nothing can distinguish causes from constants — you never see the countries with the same conditions that stayed peaceful. KKV: select on X, never on Y.

? A result is significant at $p < .001$. Is it important?

Unknown. Significance answers "is this probably not zero?" — with a large enough n , a trivial difference clears any threshold. Importance is about **effect size**. Levitt's capital punishment estimate explains about 1.5% of the 1990s homicide decline: possibly real, definitely not the answer. And significance says nothing about causality either.

? What does a 95% confidence interval mean?

That the **method** captures the true value 95% of the time in repeated sampling. It does *not* mean there's a 95% chance the truth is inside this particular interval — the truth is fixed, and this interval either contains it or doesn't. Formula: estimate $\pm (1.96 \times \text{standard error})$.

? Explain an antecedent variable versus an intervening variable.

An **antecedent** variable comes before X and causes both X and Y — that makes the X – Y relationship **spurious** and kills the claim. An **intervening** variable sits between X and Y and transmits the effect — that **explains** the claim and leaves it standing. Same-looking diagram, opposite verdict.

? In the Gerber, Green & Larimer experiment, why four treatments instead of one?

To identify the **mechanism**. A single mailer would only show that mail works. The escalating series — civic duty (+1.8), you're being studied (+2.5), your own voting record (+4.9), your neighbors' records (+8.1) — shows that the driver isn't information or reminding, it's **social surveillance**. Each step adds one ingredient and the jump tells you what that ingredient is worth.

? You have a strong correlation, $r = 0.79$, $p < .0001$. Name three ways it could still be non-causal.

Reverse causation — Y causes X . **Spuriousness** — some Z causes both. **Ecological fallacy** — the correlation exists at the group level but not the individual level. Messerli's chocolate-and-Nobel-prizes paper has all three at once, which is why it's on the syllabus.

Glossary

Antecedent variable — comes before X , causes both X and Y ; source of spuriousness.

Attrition — non-random dropout that makes groups non-comparable.

Bias — systematic error in one direction; not fixed by more data.

Central limit theorem — sample means are normally distributed even when the variable isn't.

Confounding variable (Z) — related to both X and Y ; if uncontrolled, produces a misleading association.

Counterfactual — what would have happened to the same case under the other condition.

Covariation — X and Y move together. Hurdle 3, and the easy one.

Coverage error — the sampling frame leaves out part of the population.

Ecological fallacy — inferring individual behavior from group-level data.

External validity — whether a finding generalizes beyond the study.

Falsifiability — the claim could in principle be shown wrong.

Fundamental problem of causal inference — only one potential outcome is ever observed per case.

Internal validity — whether the relationship inside the study is really causal.

Intervening variable — sits between X and Y and transmits the effect; explains rather than refutes.

Margin of error — the random sampling error only; $\approx 1/\sqrt{n}$ for 95% confidence.

Null hypothesis — the claim of no relationship; what a significance test actually evaluates.

Operational definition — the exact procedure used to measure a concept.

p-value — probability of a result this extreme if the null were true.

Parameter vs. statistic — the population's true value vs. your sample's estimate of it.

Random assignment — chance decides who is treated; buys internal validity.

Random sampling — chance decides who is studied; buys external validity.

Regression coefficient (b) — predicted change in Y per one-unit increase in X , holding other variables constant.

Regression to the mean — extreme cases drift back toward average on their own.

Reliability — the same procedure gives the same result each time.

Representativeness ratio — group's share of voters $\div$ share of eligible population; 1.0 is parity.

Sampling distribution — the distribution of a statistic across all possible samples.

Selection bias — the groups differed before any treatment; the top threat in observational work.

Spurious relationship — X and Y covary only because Z causes both.

Standard error — the standard deviation of a sampling distribution; how much an estimate bounces.

Statistical significance — the result is unlikely under the null; says nothing about size or cause.

Type I / Type II error — false positive / false negative.

Validity — you are measuring the thing you claim to measure.

Voluntary response sample — respondents select themselves in; systematically biased.

How to use this. Before each quiz, read the relevant units and try the self-test cold. Before the midterm and final, do one pass through the four hurdles and one pass through the seven case files — those two frames carry most of the points. When a new reading appears, ask the same three questions: what's the claim, what's the comparison, and what would change my mind.

Built from the Fall 2026 syllabus and the readings posted to the course site. Several of the textbook excerpts on Brightspace are image-only scans with no searchable text — the Pollock, Kellstedt & Whitten, Johnson et al., King/Keohane/Verba and Moore & Notz sections here follow the standard published editions of those chapters, so check page-specific details against your own copies. Everything drawn from the articles and the survey-based readings is taken directly from those texts.

Audio editions. Narrated with a locally-run neural voice. The recordings follow this page but are written for the ear: figures are described rather than shown, and the full edition closes with a spoken self-test and a read-aloud glossary.

Not a substitute for doing the reading — Trounstine grades on whether you can apply the ideas to a case you've never seen.